Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure your UD **email** AND **name** are below. Unless your UD emails are included in this table, you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Austin Myhre	myhreap@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.

The site will allow users to log their own lunar observations, track visibility and weather, and store notes about the observations. The purpose is to give astronomy students a place to record data and review observation patterns.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Many apps show data of all types regarding the moon; however, not many allow for personal recordings to be noted or taken into account without some kind of paywall.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I want this site to be used to help students or other astronomers log data and view trends. Having a program to help with this could have helped me on an assignment for my astronomy class.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal is to collect and store user data and help show patterns in the data.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Lunar phase, date/time, visibility (Bortlescale), sky conditions, optional notes

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Calculate average weather conditions over a time period, recall specific conditions about an observation.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

A log of all observations, visibility summaries, and sky condition patterns

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *if building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Observations, Lunar phases

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Needs: Phase name, visibility rating, sky conditions, notes, date/time Ignore: Exact coordinates, observation method, temperature, moonrise/set times

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult part will be making sure the data is stored properly and making sure the route functions come together properly.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will avoid shallow learning by reading through and understanding any code created by LLMs as the project is worked on. By going through line by line and checking the LLM's code and implementing the code myself, I can make sure I fully understand what the code is doing and how it works.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

The ultimate goal is to get a good grade. As well as learning Drafter and general coding more in-depth.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool for lunar observation tracking. What features would be useful for astronomy students, and what problems do people face when trying to log sky data

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI mentions data fragmentation as a common issue, with data coming from many different sources like a specific moon app, weather apps, and moonrise times. The app can collect the data into one place. It also mentions how field notes are inconsistent and can be lost; having them online makes it harder to lose. It suggested a DarkSky UI where everything is in high-contrast red on black for lower light levels when writing observations

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The dark sky UI is a bit ambitious for a beginner project; simplify the UI idea

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

The AI gave similar ideas to what I have already come up with; however, the ui idea is new and not a bad idea. However, I think I may simplify it to be a black background, white text

Part 6: Resource Planning

1. What resources will you use if you get stuck?

The textbook, LLM, and TA's

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Read the error message Check where the function occurred Put a print statement under the function to test the output Check spelling, indentation, and colons Last resort, ask the LLM to explain the error code and why it happens

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

The only AI used is from part 5

Part 8: Submit

In Google Docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.